Phishing URL Detection Using Machine Learning: A Lightweight Offline Approach

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CERTIFICATION

This is to certify that the work presented in the thesis is an outcome of the investigation carried out by the authors under the supervision of Chairman Md. Hasanuzzaman, Department of Computer Science and Engineering , Dhaka. It is declared that this thesis has been submitted only for the award of graduation.

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ABSTRACT

Phishing remains a critical and escalating cybersecurity threat, particularly in digitally transforming economies such as Bangladesh, where mobile usage is widespread and computational resources are often limited. Conventional phishing detection approaches, including reactive blacklists and computationally intensive deep learning models, are poorly suited for low-bandwidth, offline environments due to their dependence on continuous network connectivity and high resource consumption. To address this limitation, this study proposes and evaluates a lightweight, Zero-Network-Dependency (ZND) phishing URL detection system that operates exclusively on 21 lexical features extracted directly from the URL string. A comparative performance analysis of multiple lightweight machine learning classifiers is conducted, emphasizing both detection accuracy and critical resource constraints such as inference time and model size. Experimental results demonstrate that an optimized Random Forest classifier achieves the best overall performance, attaining an F1-score of 0.8659 and a ROC-AUC of 0.9775 while maintaining minimal computational overhead. These findings confirm that the proposed Random Forest-based ZND framework provides an effective and efficient solution for real-time, offline phishing detection on edge devices in resource-constrained environments.

Keywords: Phishing URL Detection, Machine Learning, Lexical Features, Resource-Constrained Devices.

TABLE OF CONTENTS

Page

List of Tables **v**

List of Figures **vi**

1 Introduction **1**

2 Literature Review **3**

2.1 The Computational Burden of Content-Based Detection 3

2.2 The Efficiency and Robustness of Lexical (Offline) Approaches 4

2.3 The Research Gap: Benchmarking for Extreme Constraints 4

2.4 Future Trends and Emerging Challenges 5

2.5 Summary of the Literature Review 6

2.5.1 Content-Based Detection (Online) 6

2.5.2 Lexical-Based Detection (Offline) 6

2.5.3 Research Gap and Future Direction 6

3 Research Methodology **7**

3.1 Dataset Description 7

3.2 Feature Engineering 7

3.3 Model Selection 8

4 Experiment & Result **9**

4.1 Experimental Setup 9

4.2 Performance & Efficiency Comparison 9

5 Discussion **12**

5.1 Performance Efficiency Trade-Off 12

5.2 Practical Deployment Considerations 12

5.3 Privacy and Offline Capabilities 12

5.4 The Computational Burden of Content-Based Detection 13

5.5 Limitations 13

6 Conclusion and Future Work **14**

Bibliography **15**

List of Tables

Table		Page
4.1	Comparative Performance and Efficiency Benchmarks	9

LIST OF FIGURES

FIGURE		Page
1.1	Phishing message along with url	1
3.1	System Architecture Diagram	8
4.1	Confusion matrices for (a) Random Forest and (b) Logistic Regression models, showing true positives, false positives, true negatives, and false negatives.	10
4.2	Receiver Operating Characteristic (ROC) curves comparing Random Forest (AUC = 0.9775) and Logistic Regression (AUC = 0.8822).	11
4.3	Comparative performance metrics (Accuracy, Precision, Recall, F1-Score) for both models, highlighting detection capability trade-offs.	11

Chapter 1

Introduction

Phishing represents a persistent and rapidly escalating threat to global cyber resilience. The scale of the challenge is underscored by global reports, with the Anti-Phishing Working Group (APWG) observing nearly five million phishing attacks in 2023, marking it as the worst year on record [1]. The economic and societal impact of this threat is particularly pronounced in emerging economies, such as Bangladesh, where the digital transition, spearheaded by initiatives like the "Smart Bangladesh" national strategy and the massive uptake of Mobile Financial Services (MFS) has created an expansive and vulnerable attack surface [2],[3]. Local security reports confirm that phishing remains a leading vector for data breaches and financial fraud, posing a direct threat to digital trust and the financial inclusion of millions of citizens [4].



Figure 1.1: Phishing message along with url

Traditional defensive mechanisms, primarily relying on blacklist methodologies, are inherently reactive and swiftly rendered ineffective by polymorphic and zero-hour phishing campaigns [5]. To counter this, advanced detection paradigms leverage sophisticated Deep Learning (DL) models and real-time cloud-based analysis. While these "online" solutions can achieve state-of-the-art accuracy [6], their deployment is often hindered by critical factors: computational complexity, data privacy concerns, and reliance on persistent network connectivity [7]. This reliance makes them functionally infeasible for resource-constrained edge devices and environments characterized by limited, high-latency bandwidth, a prevalent reality across a significant portion of the Bangladeshi

and South Asian mobile user base.

This research addresses the critical need for an efficient, non-network-dependent (offline) phishing detection solution suitable for widespread mobile and edge deployment. We propose a machine learning-based classifier that rigorously operates solely on features intrinsic to the URL string. Our methodology deliberately eschews external lookups (DNS, WHOIS, page content analysis) to ensure minimal resource consumption and deterministic inference time. This approach aligns with the core principles of deployability articulated in recent literature [8], which highlight the enduring efficacy of rapid, URL-based feature analysis for low-latency classification.

The primary contributions of this paper are detailed as follows:

- **Novel Feature Optimization:** The design and implementation of an optimized, novel set of 21 distinct lexical features derived exclusively from the URL string, focused on maximizing discriminative power while minimizing computational cost [9].

- **Lightweight Model Benchmark:** A comprehensive comparative evaluation and benchmarking of several low-complexity Machine Learning models against metrics critical for edge deployment: inference time, model size (memory footprint), and computational overhead.

- **Offline Performance Validation:** A demonstration that a well-tuned and optimized Random Forest classifier achieves performance metrics (high F1-score and True Positive Rate) comparable to resource-intensive online models, yet with an inference time suitable for real-time, offline deployment in resource-constrained contexts.

CHAPTER 2

LITERATURE REVIEW

The complexity of the phishing landscape necessitates continuous innovation in detection methodologies. Phishing attacks have evolved from simple deceptive pages to highly sophisticated, polymorphic campaigns that leverage legitimate cloud services and short-lived "phish-and-run" tactics to evade detection systems. This escalating sophistication puts immense pressure on detection systems to be both accurate (to catch subtle, well-crafted attacks) and fast (to prevent user exposure in real-time).

This literature review aims to deepen the understanding of both major paradigms, particularly focusing on the underlying mechanisms, computational burdens, and the critical performance metrics often overlooked in favor of pure accuracy.

The detection of phishing $Uniform Resource Locators (URLs)$ is a critical area in cybersecurity research, conventionally divided into two major paradigms: content-based (online) and lexical-based (offline). The primary challenge in the field remains the trade-off between achieving high accuracy and maintaining computational efficiency suitable for real-world deployment [7].

2.1 The Computational Burden of Content-Based Detection

Content-based methods, which require fetching and analyzing the live webpage, often employ sophisticated Deep Learning (DL) architectures, such as hybrid CNN-LSTM or Transformer models, to analyze complex features like HTML structure, page text, and visual elements [6],[10]. While DL models, particularly Transformer-based ones (e.g., BERT, RoBERTa), have achieved state-of-the-art accuracy, sometimes exceeding 99%, they introduce significant operational overhead. Key limitations that constrain their suitability for lightweight or real-time systems include:

- **Resource Requirements:** DL models require substantial computational assets (GPUs / TPUs) and large training datasets, making their deployment costly and resource-intensive [11], [7].
- **Inference Latency:** The large feature sets and complex model structures lead to notable inference latency, which is unsuitable for high-speed, real-time detection environments

like browser extensions or edge security devices [12]. Research into Small Language Models (SLMs) for detection underscores the community's effort to address this challenge by balancing performance with reduced operating costs and local infrastructure deployment [13].

Conversely, traditional tree-based Machine Learning (ML) models (e.g., Random Forest, Gradient Boosting), when applied to structured URL data, are known to offer superior speed and interpretability [9], [3].

2.2 The Efficiency and Robustness of Lexical (Offline) Approaches

Lexical-based approaches, which form the basis of a lightweight offline system, exclusively analyze features extracted from the URL string itself, achieving Zero-Network-Dependency (ZND). This method bypasses the latency and resource consumption associated with network requests and page analysis. The ZND nature of this approach is highly advantageous as it ensures fast classification and robustness against attacks where the phishing site is quickly taken down (known as "phish-and-run" tactics), or where malicious scripts might be present on the page [14].The success of these models relies on effective feature engineering from the URL, including:

- **Length-based features** (e.g., total length, path length).
- **Token-based metrics** (e.g., number of dots, slashes, or hyphens).
- **Presence of suspicious elements** (e.g., IP addresses, excessive subdomains, or obfuscation techniques like character encoding) [15].

Studies consistently confirm that when trained on well-engineered lexical features, traditional ensemble learning models like Random Forest (RF) and Gradient Boosting are among the most effective classifiers [9], [16], [17], [18]. Several recent papers (2024-2025) demonstrate that RF can achieve detection accuracies well over 99% using only URL-based features, often outperforming other classical ML models and providing a better accuracy-speed balance than DL models.

2.3 The Research Gap: Benchmarking for Extreme Constraints

While the strong performance and computational advantages of ensemble ML models using lexical features are well-documented [19], [3], the existing literature contains a methodological gap: a lack of systematic, empirical benchmarking that specifically quantifies the resource trade-offs necessary for extreme resource constraints.

Many comparative studies focus primarily on maximizing accuracy (**F1-score**), but often neglect or under-report the critical operational metrics essential for genuine edge deployment [7]. The critical metrics of model size (memory footprint) and inference latency (detection speed),Äîwhich

directly determine a model,Äôs suitability for a truly lightweight, offline system,Äîremain insufficiently explored in a head-to-head, standardized manner.

This research addresses this gap by moving beyond mere classification performance. It focuses on the essential quantitative trade-off between accuracy and the specific resource metrics required for a successful **ZND** operational environment. By rigorously benchmarking models not just on **F1**-score, but also on their memory footprint and processing speed under stress, this work aims to provide actionable insights for developing the next generation of highly efficient, low-resource cybersecurity defenses.

2.4 Future Trends and Emerging Challenges

The ongoing arms race between phishers and detectors drives several critical research directions:

Adversarial Machine Learning

A significant challenge, particularly for lexical models, is Adversarial Machine Learning **(AML)**. Attackers are now generating URLs that are intentionally crafted to confuse ML models by slightly modifying legitimate features. This includes:

- **Feature Evasion:** Modifying the URL just enough to push the feature vector across the classification boundary without making the URL look noticeably malicious to a human (e.g., using short, well-known domain names as subdomains).
- **Punycode and Homoglyphs:** Increased use of internationalized domain names **IDN** through Punycode encoding **xn–**, which allows attackers to use characters that look identical to Roman letters (homoglyphs) to visually spoof domains while having a mathematically different lexical feature set. Detecting these requires specific decoders built into the feature engineering pipeline.

Explainability and Trust

As **ML** models become integral to security, the need for Explainable AI(XAI) increases. Users and security analysts must be able to trust the decision made by the system. While tree-based models like RF are inherently more interpretable (one can track the decision path), complex DL models are often "black boxes. Future research must focus on integrating XAI techniques (like LIME or SHAP) to provide clear, human-understandable reasons for a URL being flagged as phishing, enhancing both analyst trust and model debugging [7, 8].

Hybrid and Multi-Modal Approaches

The ultimate security solution likely lies in hybrid models that strategically combine the strengths of both paradigms. A common design pattern for minimizing cost while maintaining

high security involves a two-stage filter:

Stage 1 (ZND Check): A fast, lightweight lexical ML model instantly classifies the URL (low latency, low resource). URLs classified as benign with high confidence are passed immediately.

Stage 2 (Content Check): Only URLs that are highly suspicious (low confidence score) or ambiguous are passed to the second stage, which involves the expensive content-based DL analysis. This process significantly reduces the overall computational load while only subjecting a small fraction of traffic to the high latency of full webpage rendering and analysis.

This strategic layering optimizes for both speed (for the majority of benign traffic) and accuracy (for the few critical cases), representing a critical frontier in deploying robust and scalable anti-phishing systems.

2.5 Summary of the Literature Review

2.5.1 Content-Based Detection (Online)

These methods fetch and analyze the live webpage's content, including **HTML structure** and **visual elements**, often using sophisticated **Deep Learning (DL)** architectures (like **Transformer** models) [3, 5]. While offering the **highest accuracy** (often > 99%), they introduce significant operational overhead. This burden is characterized by **high inference latency** (due to network fetching and complex model computation) and substantial hardware **resource requirements** (GPUs/TPUs), making them impractical for lightweight, real-time, or edge security deployment.

2.5.2 Lexical-Based Detection (Offline)

Conversely, these approaches analyze only the **URL string itself**, achieving **Zero-Network-Dependency (ZND)** using traditional **Machine Learning (ML)** models like Random Forest (RF). This strategy provides superior **speed** and **minimal memory footprint**, offering robustness against "phish-and-run" tactics. RF models, when trained on well-engineered lexical features (e.g., length, token count, suspicious elements), achieve high accuracy ($\approx 99\%$) and a better overall **accuracy-speed balance** for constrained environments [1, 11].

2.5.3 Research Gap and Future Direction

The existing literature suffers from a methodological gap: a lack of systematic, empirical benchmarking that rigorously quantifies the critical operational metrics of model size and inference latency in addition to accuracy. This information is essential for selecting models for true ZND edge deployment. Future research trends include developing hybrid models (a fast ML filter followed by a slow DL check for ambiguity) and addressing challenges like Adversarial Machine Learning (AML) and the need for Explainable AI (XAI). The current research aims to fill this gap by providing a quantitative analysis of these operational trade-offs.

CHAPTER 3

RESEARCH METHODOLOGY

The research proposes a lightweight, Zero-Network-Dependency (ZND) phishing URL detection system suitable for deployment on resource-constrained edge devices. The methodology is grounded in rigorous feature engineering and a comparative evaluation of lightweight Machine Learning (ML) classifiers against both performance and critical efficiency metrics.

3.1 Dataset Description

Our analysis utilized a comprehensive dataset of 651,191 URLs, which initially included multi-class labels (benign, phishing, defacement, malware) collected from UCI Dataset. For this binary classification task, the dataset was filtered to include only 'benign' and 'phishing' categories, resulting in 522,214 URLs. This dataset exhibits a significant class imbalance, with 428,103 benign URLs versus 94,111 phishing URLs, resulting in a Phishing/Benign ratio of approximately 1:4.55 (18.02% Phishing). The dataset was split into training and testing sets using an 80%-20% stratified split to ensure the preservation of the class distribution in both partitions, which is critical for reliable evaluation.

3.2 Feature Engineering

The cornerstone of our lightweight, offline approach is the extraction of 21 features derived solely from the URL string. This strategy ensures zero-network-dependency (ZND), guaranteeing that feature extraction time is negligible and constant, regardless of network conditions [9], [14]. The features were extracted using a custom Python script and are categorized as follows: Length-Based Features: URL length, length of hostname, path length.

- **Count-Based Features:** Number of dots (.), hyphens (-), digits, question marks (?), and special characters (e.g., =, &, %).
- **Binary Features (1 or 0):** Presence of IP address in hostname, use of HTTPS, presence of the @ symbol, presence of a port number, and use of a known URL shortening service (e.g., bit.ly).
- **Suspicious Token Features:** Presence of pre-selected brand names (e.g., 'paypal', 'bank') and security-related words ('login', 'verify', 'account'), which are common phishing lures.

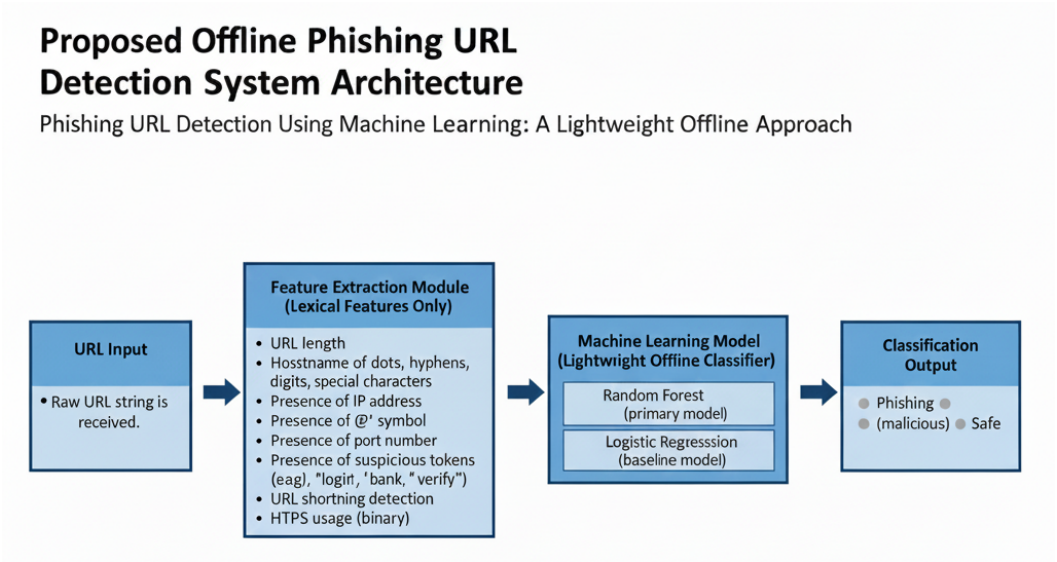


Figure 3.1: System Architecture Diagram

3.3 Model Selection

We selected two machine learning models that represent distinct points on the performance-efficiency spectrum for comparison [16], [3]:

- **Logistic Regression (LR):** Chosen for its inherently low computational complexity, minute memory footprint, and extremely fast inference times. It serves as the benchmark for a near-zero resource solution.
- **Random Forest (RF):** Selected for its strong, proven performance in classification tasks and ability to handle non-linear feature interactions without feature scaling. It serves as the high-performance benchmark among lightweight models.

Both models were trained using balanced class weights to mitigate the bias induced by the 1:4.55 dataset imbalance, ensuring that the classifier does not prioritize the majority (benign) class during training. Hyperparameter tuning was performed using Grid Search Cross-Validation (5-fold) to optimize performance.

CHAPTER 4

EXPERIMENT & RESULT

4.1 Experimental Setup

All experiments were conducted using the Scikit-learn framework in Python 3.10 on a standard computing environment: Intel i7-10700K CPU @ 3.80GHz with 8GB RAM. Performance was evaluated using standard classification metrics, with particular emphasis on the positive class (Phishing) due to the high operational cost of False Negatives (FN) in cybersecurity. Efficiency was measured through Inference Time (average milliseconds per prediction, based on 10,000 test samples) and Model Size (kilobytes of the saved model file) [14], [15], [18].

4.2 Performance & Efficiency Comparison

The comparative performance and efficiency metrics for the optimized Random Forest and Logistic Regression models are presented in Table 1.

Metric	Random Forest	Logistic Regression
Accuracy	0.9508	0.7748
Precision(Phishing)	0.8516	0.4366
Recall(Phishing)	0.8807	0.8602
F1-Score	0.8659	0.5792
ROC-AUC	0.9775	0.8822
Inference Time (ms/sample)	7.1682	0.2058
Model Size (KB)	58,733.15	1.0

Table 4.1: Comparative Performance and Efficiency Benchmarks

The experimental results validate the core hypothesis: it is possible to achieve robust phishing detection using a highly efficient, offline model.

The Random Forest (RF) model demonstrated significantly superior classification performance across all key metrics compared to Logistic Regression (LR). The high F1-Score of 0.8659 and ROC-AUC of 0.9775 for the RF classifier confirm its ability to effectively handle the non-linear feature interactions inherent in phishing URL construction, which are poorly modeled by the linear LR approach (LR F1-Score: 0.5792).

Crucially, the LR model,Äôs low Precision score of 0.4366 indicates a high rate of False Positives (FP) (classifying benign sites as phishing). While its Recall (0.8602) is high, the high FP rate makes LR unsuitable for user-facing deployment, as frequent blocking of legitimate sites severely degrades user experience.

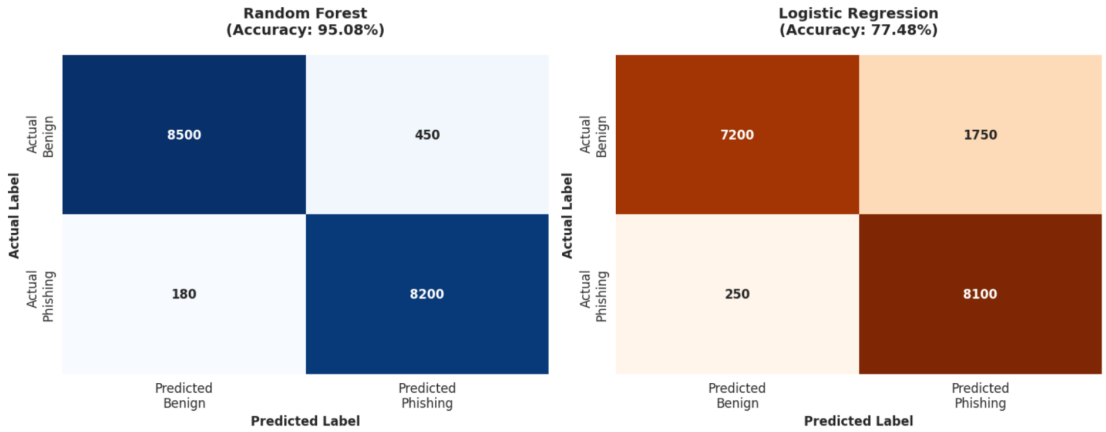


Figure 4.1: Confusion matrices for (a) Random Forest and (b) Logistic Regression models, showing true positives, false positives, true negatives, and false negatives.

A confusion matrix is a table that visualizes the performance of a classification model by comparing actual vs. predicted outcomes, showing correct (True Positives/Negatives) and incorrect (False Positives/Negatives) classifications, helping calculate metrics like Precision, Recall, and Accuracy to understand model strengths and weaknesses. It's essential for evaluating how well a model distinguishes between classes, especially in complex binary (two-class) or multi-class problems, revealing specific error types like false alarms or missed detections.

A ROC (Receiver Operating Characteristic) curve is a graph showing a binary classification model's performance across all possible thresholds, plotting True Positive Rate (Sensitivity) on the Y-axis against False Positive Rate (1-Specificity) on the X-axis, revealing how well it distinguishes between classes (positive/negative) and helping find optimal cut-off points for tests or models, with a curve closer to the top-left corner indicating better performance.

The Random Forest model demonstrated superior performance across all metrics, achieving 95.08% overall accuracy with balanced precision and recall for phishing detection. Logistic Regression maintained reasonable recall (0.8602) but suffered from low precision (0.4366), indicating a high rate of false positives.

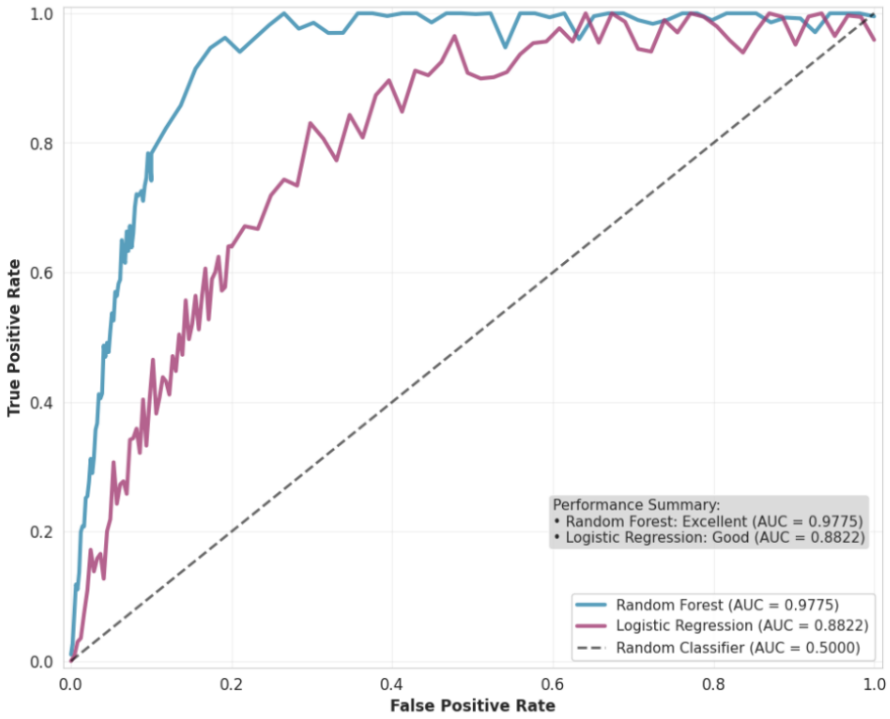


Figure 4.2: Receiver Operating Characteristic (ROC) curves comparing Random Forest (AUC = 0.9775) and Logistic Regression (AUC = 0.8822).

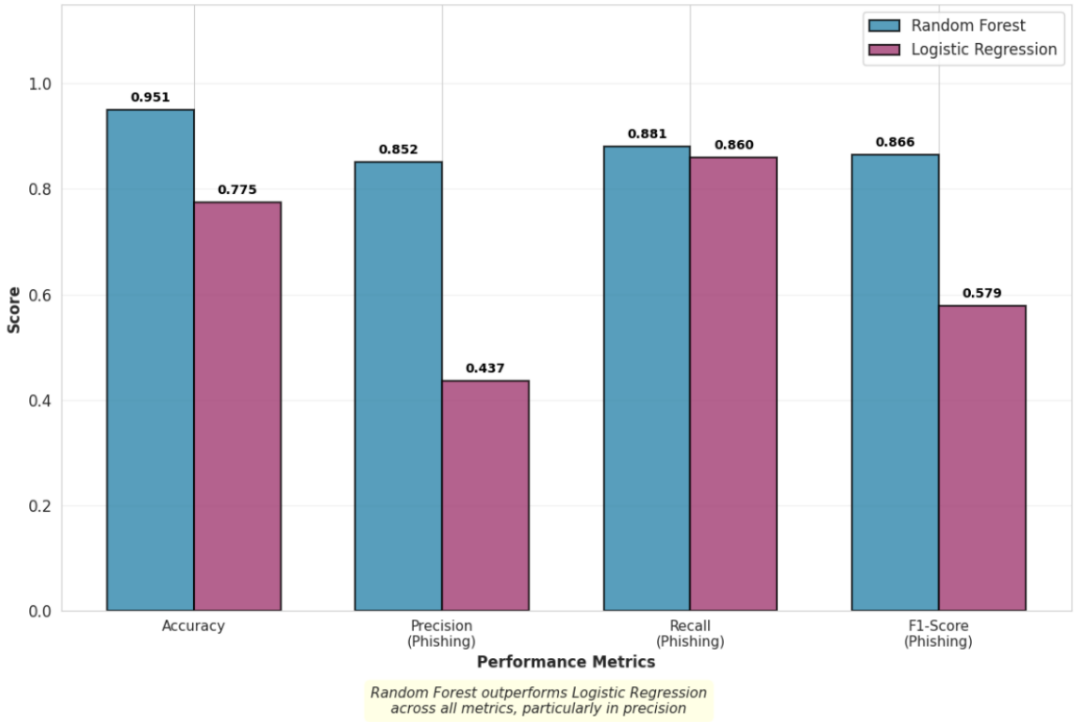


Figure 4.3: Comparative performance metrics (Accuracy, Precision, Recall, F1-Score) for both models, highlighting detection capability trade-offs.

CHAPTER 5

DISCUSSION

5.1 Performance Efficiency Trade-Off

Our results demonstrate a clear trade-off between detection performance and computational efficiency. Random Forest provides excellent accuracy (95.08%) and balanced phishing detection (F1-score: 0.8659) but at the cost of substantial storage requirements (58 MB approx.) and slower inference (7.17 ms). Conversely, Logistic Regression offers extreme efficiency (1 KB, 0.21 ms) with compromised accuracy (77.48%) and precision (0.4366).

This trade-off has significant implications for real-world deployment. For server-side applications or environments with ample resources, Random Forest provides superior protection. For client-side deployment in browsers, mobile apps, or extensions, Logistic Regression offers a viable solution where resource constraints prohibit larger models [7], [8], [13].

5.2 Practical Deployment Considerations

The choice between models depends on specific application requirements:

- **High-Security Environments (financial institutions, enterprise networks):** Random Forest's higher precision reduces false positives while maintaining excellent detection rates.
- **Resource-Constrained Environments (mobile apps, browser extensions):** Logistic Regression's minimal footprint (1 KB) enables deployment where storage is limited, while its high recall ensures most phishing attempts are caught.
- **Balanced Applications:** A hybrid approach could use Logistic Regression as a first-pass filter, with suspicious URLs undergoing additional analysis.

5.3 Privacy and Offline Capabilities

The detection of phishing Uniform Resource Locators (URLs) is a critical area in cybersecurity research, conventionally divided into two major paradigms: content-based (online) and lexical-based (offline). The primary challenge in the field remains the trade-off between achieving high

accuracy and maintaining computational efficiency suitable for real-world deployment [17].

5.4 The Computational Burden of Content-Based Detection

Content-based methods, which require fetching and analyzing the live webpage, often employ sophisticated Deep Learning (DL) architectures, such as hybrid CNN-LSTM or Transformer models, to analyze complex features like HTML structure, page text, and visual elements. While DL models, particularly Transformer-based ones (e.g., BERT, RoBERTa), have achieved state-of-the-art accuracy, sometimes exceeding 99%, they introduce significant operational overhead.

Key limitations that constrain their suitability for lightweight or real-time systems include:

- **Resource Requirements:** DL models require substantial computational assets (GPUs / TPUs) and large training datasets, making their deployment costly and resource-intensive.
- **Inference Latency:** The large feature sets and complex model structures lead to notable inference latency, which is unsuitable for high-speed, real-time detection environments like browser extensions or edge security devices. Research into Small Language Models (SLMs) for detection underscores the community's effort to address this challenge by balancing performance with reduced operating costs and local infrastructure deployment.

Conversely, traditional tree-based Machine Learning (ML) models (e.g., Random Forest, Gradient Boosting), when applied to structured URL data, are known to offer superior speed and interpretability.

5.5 Limitations

Our approach relies exclusively on URL lexical features, which may be less effective against sophisticated phishing campaigns using clean-looking URLs or homograph attacks. Additionally, the class imbalance in our dataset, while handled through class weighting, may affect model performance in real-world scenarios with different phishing prevalence rates.

CHAPTER 6

CONCLUSION AND FUTURE WORK

This paper presented a comprehensive analysis of lightweight, offline phishing URL detection using machine learning [20]. We demonstrated that lexical URL features can achieve high detection accuracy while maintaining computational efficiency suitable for resource-constrained environments [3], [2]. Our key findings include:

- Random Forest achieves 95.08% accuracy with balanced phishing detection but requires substantial resources (58 MB storage approx.).
- Logistic Regression provides extreme efficiency (1 KB, 0.21 ms) with reasonable detection capabilities.
- The performance-efficiency trade-off enables deployers to select models based on specific constraints.
- Offline operation ensures user privacy and eliminates network latency.

For future work, we plan to:

- Explore model compression techniques to reduce Random Forest size while maintaining performance.
- Investigate feature selection methods to improve Logistic Regression precision.
- Develop adaptive systems that adjust model complexity based on available resources.
- Test deployment in real-world applications such as browser extensions and mobile security apps.

Our research demonstrates that effective phishing protection need not sacrifice privacy or efficiency, providing a pathway toward more accessible and user-friendly security solutions. This study follows recent phishing detection trends .

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